
Correct Scores, Wrong Scientific Conclusions: Auditing Learned Surrogates of Chaotic Dynamics

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Abstract

1 Learned forecasters increasingly stand in for scientific models that are themselves
2 imperfect, and a forecast score alone cannot say whether a surrogate succeeds
3 for the right reasons. We test five conditions that connect a score to a scientific
4 claim (held-out generalization, complete failure accounting, matched disclosure,
5 structural identifiability, and cross-system transfer) on k -pendulum ($k \in \{1, 2, 3\}$)
6 and Lorenz-63 forecasts from language models, a time-series foundation model,
7 learned dynamics, sparse identification, and numerical integrators. Every test
8 alters or overturns a conclusion the raw comparison invites: the in-sample/out-of-
9 sample gap runs up to elevenfold, two exact symmetries make absolute parameter
10 recovery impossible, and a second system reverses the model ordering. On video-
11 tracked hardware the roles invert: the RK4 model with published constants is itself
12 imperfect, and its lead over simple baselines is gone within a second. A correct
13 leaderboard can still support incorrect scientific conclusions.

14 1 Introduction

15 Scientific claims require more than low forecast error. When a learned forecaster stands in for an
16 imperfect scientific model, a result may depend on training trajectories, dropped failures, unmatched
17 inputs, an unidentifiable target, or one favorable system: failures of the evaluation rather than of the
18 surrogate. We study these failure modes directly, predicting a chaotic system’s state s_T from its state
19 s_0 on two systems chosen to disagree: the k -link pendulum, chaotic for $k \geq 2$, with high-precision
20 ground truth and control over what physical metadata a model is told, and Lorenz-63, which differs
21 in observation model and hypothesis class; a hardware replication of the audit (§4.5) then scores the
22 reference model itself as the imperfect one.

23 Our contribution is a claim audit for dynamical-system forecasts: held-out generalization, complete
24 failure accounting, matched disclosure on identical trajectories, identifiability checked before scoring
25 recovery, and transfer to a second system. The five are indexed to the claims, not drawn from a
26 taxonomy: each tests one inference step between a score and a conclusion (sample to population,
27 answered cells to full task, correlation to attribution, accuracy to parameter recovery, one system
28 to a family claim), and claims we do not make would need other checks, such as calibration for
29 probabilistic forecasts. Table 1 states the conclusion before and after each test; all five alter or reject
30 a conclusion the unaudited comparison supports, and these are conclusions the field draws: zero-shot
31 forecasters are routinely scored without failure accounting or matched controls [Gruver et al., 2023,
32 Zhang and Gilpin, 2025]. Three checks are the familiar concerns of imperfect models applied to the
33 evaluation: misspecification (a Hamiltonian prior on damped dynamics), identifiability, and transfer.

34 Appendix D situates the audited families and background.

Table 1: Conclusions before and after the claim audit.

Test	Unaudited conclusion	Audited conclusion
Failures	Answered forecasts represent model performance.	Incomplete coverage lowers reliability-adjusted performance.
Generalization	Learned dynamics forecast accurately.	Held-out error is up to $11\times$ in-sample; simple baselines win.
Disclosure	Withholding constants improves forecasts.	Matched: small, model-dependent ($+0.069$ rad pooled).
Identifiability	Absolute constants can be recovered from angles.	Two exact symmetries make absolute recovery impossible.
Transfer	The pendulum order ranks model families.	Lorenz-63 produces a different order.

2 Benchmark design

2.1 Task and systems

The pendulum state is $\mathbf{s}_0 = (\theta_1, \dots, \theta_k, \omega_1, \dots, \omega_k) \in \mathbb{R}^{2k}$; hiding its constants does not change the trajectory, which permits the matched test in §4.2. Ground truth integrates the full nonlinear Lagrangian equations of motion with fixed-step RK4 at $\Delta t = 10^{-3}$ s, recorded every 0.01 s plus a 10 s pre- $t=0$ context for the time-series models. Initial conditions are sampled as $\theta_i \sim \mathcal{U}(-3, 3)$ rad, $\omega_i \sim \mathcal{U}(-1, 1)$ rad/s with a fixed seed; the labels come from RK4, so its coarser-step variant is a reference by construction, not a competitor.

2.2 Regimes and grid

Three regimes test generalization: normal ($g=9.81$, undamped), changed_disclosed ($g=3.71$, damped), and changed_hidden ($g=1.62$, damped, constants withheld; full values in Table 10). A cell is a tuple (model, k , regime, horizon) with horizons $\{0.01, 1, 10, 60\}$ s. The LLM grid also varies modality and prompting; all comparisons here use the coordinate modality with direct prompting.

Out-of-sample protocol. Learned models are never scored on their training data: a shared held-out set (a separate seed) of 180 initial conditions (60 per regime, $k \in \{1, 2, 3\}$) is scored identically for every family. The cross-family comparison uses horizons $\{1, 10\}$ s, where chaos has begun but error has not saturated ($180 \times 2 = 360$ cells per model); in-sample error appears once, to quantify the generalization gap (§4.1).

2.3 Inputs and grouping

Each family sees different information; the asymmetry is part of the experimental record (Table 3, Appendix B). The integrators get the exact equations and constants; the learned models get target-system training plus the constants (normal-regime priors in hidden cells); the LLMs get a text description with the constants except in hidden cells; Chronos-2 gets a 10 s pre- $t=0$ history window and no constants at all. The LLMs are large hosted models, while the learned models are small MLPs (Appendix B); larger physics-informed networks are out of scope. We separate known-physics methods (RK4, Euler, symplectic), which serve as numerical references, from the black-box methods being compared.

2.4 Metrics

The primary metric is mean absolute angle error (rad), wrapped to $[-\pi, \pi]$ and averaged over links and trajectories; angles alone are scored, since wrapping gives answers and failures one bounded scale that unbounded velocities lack. A cell is answered only if the model returns a parseable, finite prediction; because scoring answered cells alone would reward abstention on hard cells, unanswered cells receive the random-guess baseline $\pi/2$ (the expected wrapped-angle error of a uniform guess). Answering is part of the score: the task is fixed-coverage forecasting, not selective prediction.

Verification record. Each cell retains its identifiers, prompt, raw response, parser outcome, prediction, token counts, and score. Local methods are reproducible from the released checkpoints and

seeds; hosted deployments can change or be withdrawn, so for them the released per-cell responses are the record that keeps the evaluation auditable.

Disclosure gap. We define $\Delta_{\text{disc}} = \text{err}(\text{hidden}) - \text{err}(\text{disclosed})$; positive values indicate a penalty for withholding constants. Because regime contrasts confound disclosure with trajectory difficulty, §4.2 also toggles disclosure on identical trajectories.

3 Methods

LLMs. We evaluate `kimi-k2.6`, `grok-4-1-fast-reasoning`, and `deepseek-v4-pro` through an OpenAI-compatible API (hosted deployments, June 2026), with a structured prompt and JSON schema (Appendix A); each cell is one temperature-0 call with uniform budgets, no best-of- N .

Time-series foundation models. We run Chronos-2 [Ansari et al., 2024] locally in its standard history-conditioned form: a 10 s pre- $t=0$ context of all $2k$ channels forecasts to the target horizon, in univariate and multivariate modes (the paired mode comparison uses 20 trajectories per cell).

Learned dynamics. Neural ODE, HNN, and LNN share a 256-wide, 3-layer tanh MLP and integrate at inference with `scipy`; we compare derivative-matching, rollout, and mixed-window rollout objectives (Appendix B). The primary models train on nine trajectories, one per (k, regime) cell; a nested budget study trains fresh checkpoints at $\{1, 5, 20\}$ trajectories per eligible cell with three initializations (HNN: undamped cells only, since damping violates Hamiltonian conservation).

Integrators. Euler, RK4, and Störmer–Verlet (symplectic) at $\Delta t=0.01$ s using the exact equations of motion; their error is from discretization alone.

Baselines. Four baselines share the grid: persistence ($\theta_T=\theta_0$), constant velocity ($\theta_T=\theta_0+\omega_0 T$), a small-angle linearized model, and a nearest-neighbor analog forecast built from the learned models’ training data. On both held-out seeds (equal-weighted; §4.3) we also fit SINDy (degree-two, STLSQ) and NVAR (two delays, degree-two, ridge); divergent rollouts are failures at $\pi/2$.

Lorenz replication. A secondary Lorenz–63 test compares numerical references, persistence, SINDy, NVAR, a Neural ODE trained on 20 single-seed trajectories, and zero-shot Chronos-2 on five five-trajectory held-out seeds, reporting training-normalized coordinate MAE at $\{0.1, 0.5, 1.0\}$ s (Appendix C). HNN and LNN are omitted: Lorenz–63 is dissipative and non-Hamiltonian.

4 Results

4.1 Failure accounting and held-out generalization

Every model is scored on the shared held-out grid of 360 cells (180 trajectories, horizons $\{1, 10\}$ s); the Answered column of Table 2 gives the fraction with valid predictions. `grok` answers 82% and `lnn` 94%: their ranks fall when missing cells receive $\pi/2$, and a mean over answered cells alone would flatter both. Under the strict π bound only these two move (`grok` eight places down, the LNN still last), so the ordering is not an artifact of the imputation constant. We report 95% bootstrap intervals over trajectories (LLM per-call variation excluded).

Table 2 ranks the models under the access conditions of §2.3; the ranking is conditional on those inputs, this training budget, and this system (§4.4 tests transfer). The numerical integrators sit at the top; every learned-dynamics model lands closer to the $\pi/2$ baseline than to the RK4 and symplectic references. The lowest-error black-box entry, the LLM `kimi` (0.538 rad, 99% answered), has an interval that only marginally excludes the nearest baseline’s: a statement about one recorded June 2026 deployment, not about model families; the Lorenz test rejects the family reading.

Under the equalized hidden-regime run, nearest-neighbor lookup (0.725) and the linearized pendulum (0.855) beat every learned model, and persistence (1.004) beats all but HNN-rollout (0.972), though those two intervals overlap. The stable learned models score 0.09–0.63 rad on their training trajectories against 0.97–1.14 held out (Table 4), a $1.7\text{--}11\times$ gap widest for the in-sample winner, the mixed-window Neural ODE.

Table 2: Out-of-sample cross-family comparison (360 cells, horizons $\{1, 10\}$ s): reliability-adjusted mean angle error (rad), unanswered cells at $\pi/2$, 95% bootstrap CI. RK4 generates the data (reference). †: configurations first evaluated in this work; ‡: simple baselines.

#	Model	Err	95% CI	Answered	Family
–	rk4 (reference)	0.059	[0.032, 0.089]	100%	numerical
1	symplectic	0.205	[0.163, 0.251]	100%	numerical
2	euler	0.405	[0.344, 0.470]	100%	numerical
3	kimi-k2.6	0.538	[0.475, 0.602]	99%	LLM
4	chronos2-multi†	0.684	[0.617, 0.752]	100%	time-series
5	chronos2-uni†	0.697	[0.631, 0.766]	100%	time-series
6	nearest-neighbor‡	0.725	[0.657, 0.797]	100%	baseline
7	linearized‡	0.855	[0.774, 0.939]	100%	baseline
8	grok-4-1-fast-reasoning	0.881	[0.805, 0.957]	82%	LLM
9	deepseek-v4-pro	0.952	[0.881, 1.027]	100%	LLM
10	hnn-rollout	0.972	[0.898, 1.047]	100%	learned
11	persistence‡	1.004	[0.934, 1.078]	100%	baseline
12	neural-ode-rollout-mixed†	1.036	[0.967, 1.104]	100%	learned
13	hnn-rollout-mixed†	1.066	[0.990, 1.143]	100%	learned
14	neural-ode	1.110	[1.039, 1.181]	100%	learned
15	neural-ode-rollout	1.121	[1.054, 1.190]	100%	learned
16	hnn	1.138	[1.059, 1.218]	100%	learned
17	constant-velocity‡	1.319	[1.241, 1.396]	100%	baseline
18	lnn†	1.356	[1.282, 1.430]	94%	learned

117 SINDy (0.883 rad) and NVAR (1.248) appear only on the two-seed replication subset (Table 5);
118 SINDy lands between Chronos-2 and the learned models, NVAR near constant velocity.

119 4.2 Matched disclosure and identifiability

120 On the held-out set the hidden regime shows lower error than the disclosed (-0.390 rad, $p_{\text{Holm}} < 10^{-3}$,
121 unpaired) and normal (-0.524) regimes; but the regimes use different trajectories, and the hidden
122 ones are slower and less divergent. Toggling disclosure on identical trajectories for the three LLMs
123 (900 paired cells on a separate matched grid; Table 9) reverses the pooled effect: hiding the constants
124 increases error ($\Delta_{\text{disc}} = +0.069$ rad, $p_{\text{Holm}} = 0.034$), with a strong penalty at $k=1$ and a small reverse
125 effect at $k=3$; the unadjusted advantage is a difficulty confound.

126 The disclosure test also has a structural limit: the angular equations are unchanged by $(m, d) \mapsto$
127 (am, ad) and $(L, g, d) \mapsto (bL, bg, b^2d)$, so for $k=1$ only g/L and $d/(mL^2)$ are identifiable, and
128 absolute g, L, m, d cannot be scored as independently recoverable. A single-state constant-naming
129 probe mostly returns Earth-standard defaults (a prior, not recovery; transcripts released); a numerical
130 check verifies both invariances to below 10^{-13} .

131 4.3 Replication across seeds and training budgets

132 On an independent held-out seed (five trajectories per cell versus 20; Table 5), the selected ordering
133 is unchanged: numerical references lowest, Chronos-2 ahead of the learned models, HNN-rollout
134 best among them. Two seeds do not estimate seed-level uncertainty. The LNN’s low primary-seed
135 coverage was a hardware artifact: its per-step Hessian exhausted GPU memory on long rollouts. A
136 CPU rerun recovers the dropped cells ($66\% \rightarrow 94\%$; both seeds use it); the remaining failures are
137 genuine divergences, scored at $\pi/2$, and the two-seed mean stays 1.366 rad.

138 **Training-budget check.** Across three initializations and both held-out seeds, Neural ODE error
139 falls from 1.090 at one trajectory per cell to 0.861 at twenty, and normal-regime HNN error from
140 1.269 to 0.981 (Table 11). The rows cover different regimes; more data helps the Neural ODE
141 monotonically, the HNN plateaus by 15 trajectories, and neither closes the gap to known physics.

142 4.4 Cross-system test: Lorenz–63

143 We test whether the pendulum conclusions transfer to Lorenz–63, a chaotic system with a polynomial
144 vector field. They do not. The identifiability verdict flips: on the pendulum the physical constants

are not recoverable from an angle trajectory (§4.2), while on Lorenz, SINDy recovers the governing equations themselves, not merely accurate forecasts. The refit selects exactly the true sparsity pattern and returns $(\hat{\sigma}, \hat{\rho}, \hat{\beta}) = (10, 28, 8/3)$ with maximum coefficient error 2.2×10^{-13} (Appendix C); forecast error stays below 2×10^{-5} in every seed. The difference is the setting, not the audit: Lorenz-63 lies inside SINDy’s hypothesis class and is fully observed; the pendulum’s absolute constants are unidentifiable from angles under any method. Identifiability is a property of the observation model and the hypothesis class, and the check makes it explicit before recovery is scored.

The model ordering also changes. Trained on one seed and evaluated on five held-out seeds (25 trajectories), a Lorenz Neural ODE reaches normalized MAE 0.014 ± 0.008 , 0.079 ± 0.030 , and 0.277 ± 0.063 at 0.1, 0.5, and 1.0 s (mean $\pm$ SE over seeds), below both Chronos-2 modes at every horizon (multivariate 0.101 ± 0.018 , 0.691 ± 0.129 , 0.771 ± 0.097 ; univariate similar); NVAR drifts to 0.429 by 1.0 s and persistence trails (0.900; Table 5). This reverses the pendulum result, where trained models lose to simple baselines: here the system-trained model beats the zero-shot foundation model. The families again see different inputs, so this is not an architecture race; what the cross-system result rejects is the pendulum ranking as a general ordering of model families.

The reversal does not rest on a favorable seed: the Neural ODE beats both Chronos-2 modes in all 15 seed-by-horizon comparisons, and the identifiability result is not a sampling claim, since the symmetries are exact and Lorenz-63 lies in SINDy’s library by construction. The test excludes the API LLMs, HNN, and LNN (a Hamiltonian surrogate is misspecified for the dissipative flow); those omissions limit what it can establish, not what it can refute, and the Lorenz ordering is not offered as the correct one.

4.5 Real data: the reference model becomes the imperfect one

On real hardware RK4 is no longer ground truth but the imperfect model. We use an open video-tracked double pendulum [Myers et al., 2020] (three free-drop trials, 500 Hz, published point-mass constants, $\sim 0.35^\circ$ median tracking noise); Trial 1 serves only for calibration (nearest-neighbor database, SINDy fit, damping fit), and Trials 2–3 supply 120 held-out states scored as before at $\{0.1, 0.5, 1.0\}$ s (Table 6, Appendix E).

RK4 with the published constants leads at 0.1 s (0.084 rad, about $14\times$ the tracking noise) but reaches 1.25 rad by 1.0 s, indistinguishable from persistence (1.23; paired difference $+0.02$, CI $[-0.12, +0.16]$); unmodeled friction and the point-mass simplification consume its advantage within a second. The fitted damping trades horizons (0.89 vs. 0.82 at 0.5 s, 1.15 vs. 1.25 at 1.0 s; the 1.0 s gain is significant, CI $[-0.20, -0.00]$). SINDy, exact on Lorenz, sits at baseline level here (1.54); the hardware’s dynamics are not in its library. Chronos-2 answers 89% of cells: states shortly after the drop lack its 5 s context (Appendix E), and failure accounting scores them at $\pi/2$. Every check transfers, and the ordering differs from both synthetic orderings.

4.6 Diagnostics

Appendix F reports the rest; two matter most: the physical-fidelity check localizes a misspecification (a symplectic HNN rollout leaves energy drift unchanged: the fault is the learned Hamiltonian, not the integrator), and every black-box family saturates near the chaotic noise floor by 1 s, hence the $\{1, 10\}$ s window. Error rises with k for all but one model; the hidden regime is easier for most models (the §4.2 confound); rollout loss is not uniformly better; and Chronos-2’s two modes are indistinguishable ($n=720$).

5 Discussion and conclusion

Performance depends on system, observations, hypothesis class, data, and access, which a scalar leaderboard does not encode: each of the five checks changes a claim it supports, so benchmarks should report checks, training budgets, and information access with the result. The synthetic systems make each check’s effect visible against exact ground truth; the hardware check runs the audit where the reference model itself is imperfect; all records to repeat it accompany the paper. **Limitations.** The LLM grid uses one held-out seed; the Lorenz test has five trajectories per seed, the hardware check three trials of one apparatus, SINDy needs the true degree in its library, and foundation-model corpora may contain pendulum-like data.

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A Prompt templates

The prompts used for every LLM evaluation cell are reproduced below verbatim.

Table 3: Inputs available to each family. State is at $t=0$; history is pre- $t=0$. Chronos-2 is history-conditioned. *In changed_hidden cells the learned models receive normal-regime prior constants instead of the true ones.

Family	State	History	Constants	Equations
Numerical (RK4/Euler/sympl.)	✓	—	✓ (all regimes)	✓ (exact)
Learned (NODE/HNN/LNN)	✓	—	✓ base; withheld*	— (learned)
Time-series (Chronos-2)	—	✓ (10 s context)	—	—
LLMs (kimi/grok/deepseek)	✓	—	✓ except changed_hidden	—

242 System prompt (direct / no-CoT).

243 You are a careful physicist predicting the future state of a k -pendulum (a chain of
 244 k point masses on massless rods, hinged in a planar chain, swinging under gravity
 245 with viscous damping). Respond with ONLY a single JSON object on one line, no prose,
 246 no code fences. Schema:
 247 {"theta": [...k floats...], "omega": [...k floats...]}
 248 theta_i is in radians (link i measured from straight-down,
 249 positive counter-clockwise)
 250 and omega_i is angular velocity in rad/s.

251 **System prompt (chain-of-thought).** Same preamble, then: “Reason step-by-step about the dynamics,
 252 then output your final answer. Wrap the final answer in `<answer>...</answer>` tags containing ONLY a
 253 single JSON object: {"theta": [...k floats...], "omega": [...k floats...]}.”

254 User prompt (coordinate modality).

255 Number of links $k = \{k\}$
 256 Physical constants:
 257 g (gravity, m/s^2): $\{g\}$
 258 L (rod lengths, m): $\{L\}$
 259 m (point masses, kg): $\{m\}$
 260 damping (viscous coefficient on omega): $\{\text{damping}\}$
 261 Initial state at $t=0$:
 262 theta_0 (rad): $\{\text{theta}_0\}$
 263 omega_0 (rad/s): $\{\text{omega}_0\}$
 264 Predict the state at $t = \{T\}$ seconds.
 265 Return theta and omega as length- $\{k\}$ JSON arrays.

266 **Hidden-constants regime.** The constants block above is replaced by:

267 Physical constants: HIDDEN. The numerical values of gravity, rod lengths, point
 268 masses, and damping are NOT disclosed. You must infer dynamics from the initial
 269 conditions alone.

270 **Parsing and scoring.** Each response is stripped of code fences and (for CoT) of everything outside the
 271 `<answer>...</answer>` tags; the first $\{\dots\}$ object is JSON-parsed and validated to contain length- k theta
 272 and omega arrays. A response that fails any check is counted as unanswered and scored at $\pi/2$.

273 B Training and compute

274 All primary learned models use a 256-wide, 3-layer tanh MLP, Adam, and a cosine-annealed learning rate,
 275 and train on a single 8 GB consumer GPU. Mixed- k rollout adds negligible parameters; its cost is wall-clock
 276 time (the HNN mixed-window run is the most expensive at ~ 6.7 h because each unroll step recomputes ∇H
 277 via autograd). Chronos-2 runs at inference only. The dataset has 67,491 samples across $k \in \{1, 2, 3\}$ and
 278 three regimes, subsampled at $\Delta t = 0.01$ s from the 10^{-3} s ground truth. Despite the sample count, the training
 279 set contains only *nine* trajectories, one for each (k, regime) cell. The budget study uses nested subsets of 1,
 280 5, and 20 trajectories per eligible cell and three training initializations. Neural ODE therefore sees 9, 45, or
 281 180 trajectories; HNN sees 3, 15, or 60 undamped trajectories. Evaluation uses the same two held-out seeds
 282 throughout.

Table 4: In-sample versus held-out error (rad) for the six stable learned models: mean over the nine training trajectories versus the shared held-out grid, both at horizons $\{1, 10\}$ s with the harness inference path. The in-sample best is the most overfit.

Model	In-sample	Held-out	Ratio
neural-ode	0.625	1.110	1.8
neural-ode-rollout	0.382	1.121	2.9
neural-ode-rollout-mixed	0.092	1.036	11.3
hnn	0.568	1.138	2.0
hnn-rollout	0.560	0.972	1.7
hnn-rollout-mixed	0.522	1.066	2.0

Training objectives. Derivative matching minimizes MSE on finite-difference $\dot{s}$; rollout loss minimizes MSE on the state after w Euler steps with gradients through the unroll; mixed-window rollout trains on several window lengths jointly (Neural ODE: $w \in \{10, 50\}$; HNN: $w \in \{5, 20\}$). The LNN’s loss requires second-order automatic differentiation; the LNN trains at full scale (67k samples, hidden 256) on a CUDA GPU. Chronos-2 resamples the context to at most 512 points and forecasts at most 64 steps.

Reproducibility records. The anonymized artifact accompanying this submission contains the code, prompts, trained weights, held-out specifications, and the per-cell records for the main held-out comparison (2,160 records), the matched disclosure test (1,800), the modality and prompting grid, and the Lorenz sweep; the two large sweeps, the training-budget and balanced-seed studies, are packaged as separate archives. All dataset and held-out seeds are fixed and recorded in the released configuration. LLMs are sampled once at temperature 0. The reported hosted-model numbers are recomputed from the saved records; re-calling the API is a new evaluation rather than a reproduction. Invalid cells are scored at $\pi/2$.

C Seed and system replication

The Lorenz-63 test uses $(\sigma, \rho, \beta) = (10, 28, 8/3)$, RK4 ground truth at $\Delta t=0.01$ s, a 5 s context, one fixed training seed, and five fixed held-out seeds (all recorded in the released configuration). Each held-out seed contains five trajectories, giving 25 held-out trajectories in total. The Neural ODE is a two-hidden-layer, width-128 tanh network trained for 300 epochs by analytic derivative matching. SINDy uses a degree-two polynomial library with STLSQ, fit to analytic derivatives at the sampled training states; the refit selects the true sparsity pattern exactly and recovers $(\hat{\sigma}, \hat{\rho}, \hat{\beta}) = (10, 28, 8/3)$ with maximum absolute coefficient error 2.2×10^{-13} . NVAR uses two delays, degree-two features, and ridge regression. Chronos-2 uses `amazon/chronos-2` in univariate and multivariate modes.

D Related work

The audited models span learned dynamics [Chen et al., 2018, Greydanus et al., 2019, Cranmer et al., 2020] on chaotic testbeds [Gilpin, 2021], zero-shot LLM and foundation-model forecasting [Gruver et al., 2023, Zhang and Gilpin, 2025, Ansari et al., 2024, Das et al., 2024, Woo et al., 2024], LLM physical reasoning [Lewkowycz et al., 2022, Wei et al., 2022, Xu et al., 2025], and sparse identification [Brunton et al., 2016, Gauthier et al., 2021]. The identifiability check draws on structural identifiability under parameter symmetries [Borgqvist et al., 2024]. Unlike in selective prediction, which reports risk together with coverage [Geifman and El-Yaniv, 2017], failed cells here stay in the main score because models are compared at fixed task coverage.

E Real-data check details

The hardware dataset [Myers et al., 2020] is openly released via Mendeley Data (CC BY 4.0 listing; the deposit includes a GPLv3 license file) with video tracking of both link angles at 500 Hz and per-frame uncertainty estimates (median 0.20° upper link, 0.43° lower). We use the published simplified point-mass parameters ($L = [0.172, 0.143]$ m, $m = [0.311, 0.111]$ kg, $g = 9.81$ m/s²). Angular velocities come from a Savitzky-Golay filter (window 42 ms, cubic, analytic derivative). The recordings begin with the pendulum positioned by hand, so the free drop is detected as the first time either link exceeds 4 rad/s; evaluation states are drawn evenly from 0.2 to 24 s after the drop, 60 per trial. The tracker zeroes its angle at the first frame, leaving a gravity-referenced offset per link; every trial ends hanging at rest with tracked angles at exact multiples of 360° (within 0.08°), which fixes the offset at zero independently of any model. A 20 ms physics-tracking check agrees, with error of 0.005–0.007 rad against a 0.006 rad tracking floor. A grid search over $\{0, 5 \times 10^{-4}, \dots, 2 \times 10^{-2}\}$ selects damping 10^{-3} on Trial 1 at the 0.5 s horizon. SINDy uses a degree-two polynomial plus first-order

Table 5: Replication results. Top: reliability-adjusted pendulum error (rad) by held-out seed and their seed mean; horizons $\{1, 10\}$ s. The primary seed has 20 trajectories per (k, regime) cell and the new seed has five. Equal-seed answer rates are 97% for LNN, 61% for SINDy, and 46% for NVAR; all other rows answer 100%. Bottom: Lorenz-63 normalized MAE, averaged within each of five held-out seeds (five trajectories each) and then across seeds, with the standard error over seeds. SINDy is below 2×10^{-5} in every cell. Lower is better.

Pendulum model	Primary seed	Second seed	Seed mean
rk4 (reference)	0.059	0.074	0.067
symplectic	0.205	0.194	0.200
euler	0.405	0.415	0.410
chronos2-uni	0.697	0.604	0.650
chronos2-multi	0.684	0.621	0.653
sindy	0.860	0.906	0.883
hnn-rollout	0.972	0.858	0.915
neural-ode-rollout-mixed	1.036	0.989	1.012
hnn-rollout-mixed	1.066	1.037	1.051
nvar	1.245	1.251	1.248
lnn	1.356	1.376	1.366
Lorenz model	0.1 s	0.5 s	1.0 s
rk4 (reference)	0.000	0.000	0.000
sindy	$<10^{-5}$	$<10^{-5}$	$<2 \times 10^{-5}$
neural-ode	0.014 (0.008)	0.079 (0.030)	0.277 (0.063)
nvar	0.012 (0.001)	0.262 (0.050)	0.429 (0.084)
chronos2-uni	0.126 (0.027)	0.700 (0.102)	0.732 (0.085)
chronos2-multi	0.101 (0.018)	0.691 (0.129)	0.771 (0.097)
euler	0.028 (0.005)	0.216 (0.023)	0.885 (0.120)
persistence	0.524 (0.054)	1.028 (0.123)	0.900 (0.057)

Table 6: Real-data check: mean wrapped angle error (rad) over both links on Trials 2–3 (120 held-out states), failures at $\pi/2$. The physics rows use the published constants; “damped” adds the Trial-1-fitted viscous coefficient. Tracking noise floor ~ 0.006 rad.

Model	0.1 s	0.5 s	1.0 s	Answered
physics-rk4	0.084	0.820	1.248	100%
physics-rk4-damped	0.086	0.892	1.145	100%
nearest-neighbor (cross-trial)	0.371	0.849	1.117	100%
chronos2	0.611	1.160	1.223	89%
constant-velocity	0.345	1.402	1.576	100%
sindy-real	0.490	1.501	1.535	99%
persistence	0.934	1.654	1.230	100%

Fourier library on Trial 1 with Savitzky–Golay derivatives; Chronos-2 receives a 5 s context resampled to 50 Hz and forecasts at most 50 steps. Failures are scored at $\pi/2$; Chronos-2’s are input-infeasible cells (insufficient context) rather than divergences, and the fixed-coverage rule scores both alike. The tracking dataset has been public since 2020 and may appear in foundation-model corpora; contamination would inflate Chronos-2’s score, so its baseline-level result is, if anything, conservative. Paired bootstrap differences over the 120 states at 1.0 s: RK4 minus persistence +0.018 (95% CI $[-0.121, +0.155]$); damped minus undamped RK4 -0.103 ($[-0.202, -0.003]$), so the damping gain at the longest horizon is significant while the RK4–persistence gap is not. States sampled along a trajectory are temporally dependent, so we repeat both with a moving-block bootstrap within each trial (blocks of 10 consecutive states, ~ 4 s): the conclusions are unchanged (damped minus RK4 $[-0.179, -0.019]$; RK4 minus persistence $[-0.131, +0.064]$), and remain so for block lengths from 2 to 12 s.

F Extended diagnostics

This appendix expands the diagnostics summarized in §4.6.

Behavior by horizon. Out-of-sample error by horizon appears in Table 14. At 1 s the ordering follows the main comparison: the integrators are near-exact, kimi (0.457) has the lowest black-box error, and the learned models are at 0.86–1.25. At 10 s all models are in the chaotic regime, and the ordering is stable: kimi (0.619) remains lowest among black-box methods while the learned models reach 1.09–1.46. The mixed-window Neural ODE is intermediate at both horizons (0.951/1.120). Figure 1 shows the growth curves over the $\{0.01, 1, 10, 60\}$ s ladder: the numerical integrators stay near zero until ~ 10 s, whereas every black-box family saturates near the chaotic noise floor by 1 s. The $\{1, 10\}$ s window used for the comparison is the range over which the families remain separable.

Breakdown by k and regime. Tables 7 and 8 report out-of-sample error by number of links and by regime. Error increases with k for every model except the mixed-window Neural ODE, which dips at $k=3$ (for kimi, $0.11 \rightarrow 0.55 \rightarrow 0.95$; the integrators are near-exact at $k=1$ and degrade as chaos sets in), and the cross-family ordering holds within each k . The `changed_hidden` regime has the lowest error for most models (for kimi, 0.46 hidden vs. 0.64 normal). For the learned models, withholding the constants reduces but does not remove the difficulty effect: hidden-regime errors are now 0.71–1.19; the earlier values favor models given the true constants. The low-gravity hidden constants ($g=1.62$) produce slower, less-divergent motion over a fixed horizon, so the unadjusted hidden-regime advantage is a difficulty confound (§4.2). A cluster bootstrap over complete held-out trajectories changes no confidence interval by more than ~ 0.03 rad and leaves the ranking unchanged.

Training-objective check. Table 12 reports the effect of the loss with architecture fixed. Under the equalized hidden-regime run, rollout loss is not uniformly better: mixed-window rollout is lower for the Neural ODE, while single- k rollout is lower for the HNN. Full-scale CUDA training of the LNN gives the highest learned-model error (1.356), with 6% of cells diverging or exceeding the rollout timeout after memory-safe recovery. We make no architecture-wide claim about LNNs; under our implementation and protocol the LNN is numerically unstable and generalizes poorly out of sample.

Physical-fidelity check. Hamiltonian and Lagrangian networks target structure preservation, so we evaluate energy drift on the undamped (`normal`) regime, where true energy is constant. Table 13 reports mean absolute drift over 10 s, normalized by $E^* = g \sum_i m_i L_i$, and the fraction of bounded rollouts. The learned models do not match numerical integrators on this axis: the lowest-drift learned model is 0.779, compared with 0.066 for the symplectic integrator and 0.494 for Euler. Symplectic rollout of the HNN leaves drift nearly unchanged, so changing the integrator does not account for the HNN’s drift. At the derivative-matching objective, HNN drift is comparable to Neural ODE drift (4.27 vs. 4.35); the LNN is unstable, with 27% bounded rollouts. These results use the same nine-trajectory learned-model budget as the main comparison.

Multivariate time-series check. On 720 paired samples, Chronos-2 multivariate and univariate errors are indistinguishable: 0.543 (95% CI [0.498, 0.589]) versus 0.549 ([0.505, 0.593]). No effect remains after Holm correction, so the larger study rejects the apparent per- k trend in the primary grid.

Predictability horizon. Pointwise error past the Lyapunov time saturates to a system-dependent constant for every model, so long-horizon error rankings carry little information. We define the predictability horizon as the longest horizon a model tracks contiguously below a threshold τ from $t=0$; the contiguous rule is used because, after divergence, a later horizon can fall below τ by angle-wrap coincidence. Table 15 sweeps τ .

G Breakdowns by link count and regime

Reliability-adjusted mean angle error (rad) on the shared held-out grid (horizons $\{1, 10\}$ s), by number of links (Table 7) and by regime (Table 8); models in the order of Table 2. Cells a model never produced (e.g. the LNN’s diverged $k=3$) are scored at $\pi/2$. Each row uses the same grid as Table 2.

H Additional result tables

Full per-model values for the horizon breakdown and the predictability horizon (both defined in Appendix F), together with the regime definitions, the matched disclosure gap, and the training-budget, training-objective, and physical-fidelity studies. The corresponding summary statistics appear in the main text.

Table 7: Out-of-sample error by number of links k .

Model	$k=1$	$k=2$	$k=3$
rk4 (reference)	0.000	0.068	0.109
symplectic	0.000	0.253	0.363
euler	0.271	0.410	0.534
kimi-k2.6	0.112	0.553	0.949
chronos2-multi	0.263	0.798	0.992
chronos2-uni	0.279	0.813	0.999
grok-4-1-fast-reasoning	0.366	1.030	1.247
deepseek-v4-pro	0.872	0.949	1.036
hnn-rollout	0.524	0.985	1.406
neural-ode-rollout-mixed	0.845	1.178	1.084
hnn-rollout-mixed	0.550	1.162	1.487
neural-ode	0.626	1.304	1.400
neural-ode-rollout	0.968	1.172	1.224
hnn	0.511	1.337	1.567
lnn	1.072	1.440	1.556

Table 8: Out-of-sample error by regime. Learned-model hidden cells use the equalized hidden-regime run. The hidden regime remains easier for most models because its low-gravity constants give slower, less-divergent motion. The unadjusted regime contrast in §4.2 consequently mixes disclosure with difficulty.

Model	normal	disclosed	hidden
rk4 (reference)	0.174	0.003	0.000
symplectic	0.387	0.177	0.051
euler	0.743	0.387	0.085
kimi-k2.6	0.635	0.519	0.461
chronos2-multi	0.818	0.826	0.409
chronos2-uni	0.818	0.798	0.475
grok-4-1-fast-reasoning	1.096	0.959	0.588
deepseek-v4-pro	1.240	1.097	0.520
hnn-rollout	1.087	1.115	0.713
neural-ode-rollout-mixed	1.260	0.836	1.011
hnn-rollout-mixed	1.244	1.172	0.782
neural-ode	1.221	1.095	1.014
neural-ode-rollout	1.281	0.894	1.189
hnn	1.262	1.303	0.850
lnn	1.525	1.416	1.127

Table 9: Matched disclosure gap (rad), paired on identical trajectories. $\Delta_{\text{disc}} = \text{err}(\text{hidden}) - \text{err}(\text{disclosed})$; positive values indicate higher error without the constants. Reliability-adjusted; paired Wilcoxon, Holm-corrected. This experiment uses its own matched grid (3 models $\times$ 2 horizons $\times$ 150 trajectories = 900 pairs), not the shared held-out set, so its absolute errors are not comparable to those in Table 8. Per-model Δ values are computed before rounding.

Model	Disclosed	Hidden	Δ_{disc}
kimi-k2.6	0.746	0.957	+0.211
grok-4-1-fast-reasoning	0.987	1.025	+0.038
deepseek-v4-pro	1.425	1.382	-0.042
pooled (paired, $n=900$)			+0.069*
$k=1$			+0.297***
$k=3$			-0.055*

* $p_{\text{Holm}} < 0.05$, *** $p_{\text{Holm}} < 10^{-3}$.

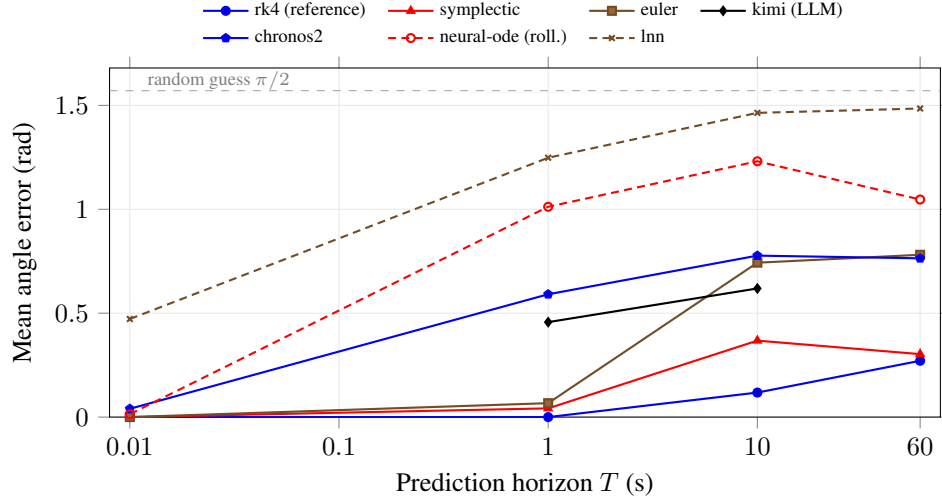


Figure 1: Error growth with prediction horizon (out-of-sample, reliability-adjusted; log time axis). The numerical integrators stay near zero until ~ 10 s; every black-box family saturates near the chaotic noise floor by 1 s. LLMs are evaluated at $\{1, 10\}$ s only. Dashed line: random-guess baseline $\pi/2$.

Table 10: Physical regimes. The changed_hidden regime withholds the constants.

Regime	g (m/s ²)	L (m)	m (kg)	damping	disclosed
normal	9.81	1.0	1.0	0.0	—
changed_disclosed	3.71	1.4/0.7/1.1	2.0/0.6/1.5	0.05	yes
changed_hidden	1.62	0.8/1.2/0.9	0.5/1.8/1.0	0.10	no

Table 11: Nested training-budget study. Reliability-adjusted pendulum error (rad), averaged over three training initializations and two held-out seeds. Intervals come from hierarchical resampling over initializations and held-out seeds. HNN is evaluated on the undamped normal regime only.

Model	Scope	Total trajectories	Error	95% CI
HNN	normal	3	1.269	[1.209, 1.349]
HNN	normal	15	0.974	[0.891, 1.062]
HNN	normal	60	0.981	[0.856, 1.113]
Neural ODE	all regimes	9	1.090	[1.011, 1.207]
Neural ODE	all regimes	45	0.968	[0.923, 1.028]
Neural ODE	all regimes	180	0.861	[0.829, 0.907]

Table 12: Training-objective study, out-of-sample mean angle error (rad) on the shared held-out grid (horizons $\{1, 10\}$ s); architecture fixed, loss/window varied. Significance from a powered paired study on the full four-horizon held-out set ($n=720$, Wilcoxon+Holm).

Model	Objective	Err	vs. prev. row
Neural ODE	derivative MSE	1.110	—
	rollout (single- k)	1.121	n.s. ($p_{\text{Holm}}=0.188$)
	mixed-window	1.036	sig. ($p_{\text{Holm}}=0.013$)
HNN	derivative MSE	1.138	—
	rollout (single- k)	0.972	sig. ($p_{\text{Holm}}=1.95 \times 10^{-4}$)
	mixed-window	1.066	sig. higher ($p_{\text{Holm}}=0.014$)

Table 13: Physical fidelity on the undamped (normal) regime: mean absolute energy drift over a 10 s rollout, normalized by $E^* = g \sum_i m_i L_i$ (0 = perfect conservation), and fraction of bounded rollouts. HNN drift is nearly unchanged when its rollout uses a symplectic integrator.

Model	Family	Energy drift	Stable
rk4	numerical	0.000	100%
symplectic	numerical	0.066	100%
euler	numerical	0.494	100%
neural-ode-rollout-mixed	learned	0.779	100%
hnn-rollout-mixed	learned	0.865	100%
neural-ode-rollout	learned	0.872	100%
hnn-rollout	learned	2.065	100%
lnn	learned	4.038	27%
hnn	learned	4.270	100%
neural-ode	learned	4.346	100%

Table 14: Out-of-sample mean angle error (rad) by horizon, reliability-adjusted, on the shared held-out grid (selected models). The cross-family comparison uses horizons $\{1, 10\}$ s.

Model	1 s	10 s
rk4 (reference)	0.000	0.118
symplectic	0.042	0.368
euler	0.067	0.743
kimi-k2.6	0.457	0.619
chronos2-multi	0.591	0.777
chronos2-uni	0.622	0.773
grok-4-1-fast-reasoning	0.905	0.857
deepseek-v4-pro	0.996	0.909
hnn-rollout	0.855	1.088
neural-ode-rollout-mixed	0.951	1.120
hnn-rollout-mixed	0.982	1.150
neural-ode	0.892	1.328
neural-ode-rollout	1.012	1.231
hnn	1.052	1.224
lnn	1.248	1.464

Table 15: Predictability horizon (s): longest horizon tracked contiguously below threshold τ , on the shared out-of-sample set (ladder $\{0.01, 1, 10, 60\}$ s). Higher is better; “—” means not even the first horizon stays below τ .

Model	$\tau=0.1$	0.25	0.5	1.0
rk4 (reference)	1.0	10	60	60
symplectic	1.0	1.0	60	60
euler	1.0	1.0	1.0	60
neural-ode-rollout	0.01	0.01	0.01	0.01
neural-ode-rollout-mixed	0.01	0.01	0.01	1.0
chronos2-multi	0.01	0.01	0.01	60
chronos2-uni	0.01	0.01	0.01	60
hnn-rollout	0.01	0.01	0.01	1.0
neural-ode	0.01	0.01	0.01	1.0
hnn	0.01	0.01	0.01	0.01
lnn	—	—	0.01	0.01